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A palladium catalyzed efficient synthesis of γ-methylene-α,β-unsaturated γ-lactones via cyclization of 3,4-alkadienoic acids

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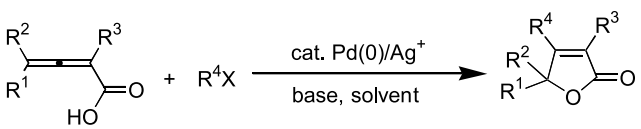
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Abstract—An efficient method was developed for the synthesis of γ-methylene-α,β-unsaturated γ-lactones from the Pd-catalyzed cyclization of 3,4-alkadienoic acids. Control experiment shows that the reaction should be carried out under a N_2 atmosphere to ensure the high purity of the products.
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1. Introduction

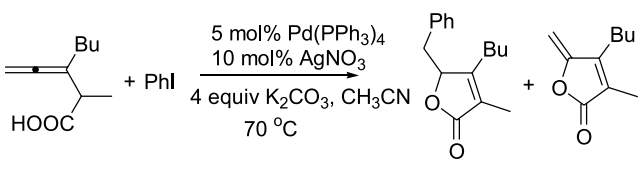
Recently, the chemistry of allenes has been attracting the attention of more and more chemists.[1,2] During the course of our studies on the chemistry of allenes,[3,4] we have developed some new methodologies for the synthesis of butenolides from α-allenoic acids (Scheme 1).[5]



Scheme 1.

2. Results and discussion

When we turned our attention from the α-allenoic acids to the β-allenoic acids **1a**, we found that under the catalysis of $Pd(0)/Ag^+$ the reaction of 2-methyl-3-(*n*-butyl)-3,4-pentadienoic acid with iodobenzene in MeCN afforded not only the coupling-cyclization product **2a** but also 5-methylene-2(5*H*)-furanone **3a** (Scheme 2). Recently, the γ-alkylidene-α,β-unsaturated γ-lactones[6] has also attracted the attention of many organic chemists because of their biological activities[7,8] and versatile use in organic synthesis.[9] When we found product **3a** was obtained in 29% yield in DMF under the catalysis of $Pd(PPh_3)_4$ (entry 1, Table 1), we went on to search a set of reaction conditions for the sole formation of **3a**. When 4 equiv $CuCl_2$ were used instead of PhI, 5-methylene-2(5*H*)-furanone **3a** was afforded under the catalysis of $PdCl_2(PPh_3)_2$ (entry 2, Table 1). However, the product was contaminated by other inseparable by-products. Furthermore, it is lucky for us to observe that when the reaction was conducted under a N_2 atmosphere, **3a** was isolated as the single product in 88% yield with a very high purity (entry 3, Table 1). When $CuBr_2$ was used instead of $CuCl_2$, the reaction was complicated (entry 4, Table 1). In the presence of H_2O or O_2, the purity of the product was also not high (entries 2, 5, and 6, Table 1). When a less amount of $CuCl_2$ was applied, the yield of **3a** was obviously lower (entries 7 and 8, Table 1). Thus, after screening several commonly used catalyst, $PdCl_2(PPh_3)_2$ appeared to be the best (compare entries 9–11 with entry 3, Table 1). The reaction could proceed in the absence of $PdCl_2(PPh_3)_2$ albeit in much lower yield (entry 12, Table 1). Obviously this cyclization reaction can also be mediated by $CuCl_2$ (entry 13, Table 1). In the absence of the base the yield was also low (entry 14, Table 1). When only 3 equiv of K_2CO_3 were added, the product isolated was not pure (entry 15, Table 1). When 5 equiv of $CuCl_2$ were applied, no obvious improvement was observed (entry 16, Table 1). From these results we concluded the best reaction conditions:

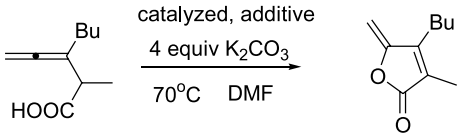


Scheme 2.

Keywords: 3,4-Alkadienoic acids; Palladium; $CuCl_2$; Cyclization.
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Table 1. Pd-catalyzed cyclization reaction of 2-methyl-3-(*n*-butyl)penta-3,4-dienoic acid **1a**



1a **3a**

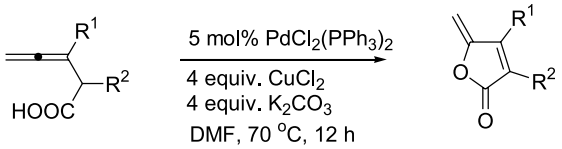
Entry	Catalyst (5 mol%)	Additive (equiv)	Time (h)	Yield of **3a** (%)
1	$Pd(PPh_3)_4$	PhI (1.5)	24	29
2[a]	$PdCl_2(PPh_3)_2$	$CuCl_2$ (4)	12	<87[b]
3	$PdCl_2(PPh_3)_2$	$CuCl_2$ (4)	12	88
4	$PdCl_2(PPh_3)_2$	$CuBr_2$ (5)	24	complicated
5[c]	$PdCl_2(PPh_3)_2$	$CuCl_2$ (4)	12	<50[b]
6[d]	$PdCl_2(PPh_3)_2$	$CuCl_2$ (4)	12	<70[b]
7	$PdCl_2(PPh_3)_2$	$CuCl_2$ (2.5)	24	75
8	$PdCl_2(PPh_3)_2$	$CuCl_2$ (0.8)	24	22
9	$PdCl_2$	$CuCl_2$ (4)	24	54
10	$Pd(OAc)_2$	$CuCl_2$ (4)	24	40
11	$PdCl_2(CH_3CN)_2$	$CuCl_2$ (4)	24	34
12	—	$CuCl_2$ (4)	24	60
13	—	$CuCl_2$ (1)	12	18
14[e]	$PdCl_2(PPh_3)_2$	$CuCl_2$ (4)	12	70
15[f]	$PdCl_2(PPh_3)_2$	$CuCl_2$ (4)	12	<87[b]
16[g]	$PdCl_2(PPh_3)_2$	$CuCl_2$ (4)	12	89

[a] The reaction was exposed to air.
[b] The product is not pure.
[c] The reaction was conducted under 1 atm of pure O_2.
[d] Four equivalents H_2O was added.
[e] No K_2CO_3 was used.
[f] Three equivalents K_2CO_3 were used.
[g] Five equivalents K_2CO_3 were used.

5 mol% $PdCl_2(PPh_3)_2$, 4 equiv of $CuCl_2$, 4 equiv of K_2CO_3 in DMF at 70 °C for 12 h under N_2.

β-Allenoic acids can be conveniently prepared from the hydrolysis of the corresponding β-allenoic acid esters.[10] Subsequently, we studied the scope of this cyclization reaction of β-allenoic acids (Table 2). From the data listed in Table 2, it is obvious that the scope of the reactant is broad: R^1 can be H, alkyl, allyl, Bn or *t*-Bu and R^2 can be H or alkyl. With simple unsubstituted pentadienoic acid, the yield of **3k** (protoanemonin)[7] is 31% by NMR (entry 11, Table 2).

Table 2. $PdCl_2(PPh_3)_2$-catalyzed cyclization reaction of β-allenoic acids **1**



1 **3**

Entry	R^1	R^2		Yield of **3** (%)
1	*n*-Bu	Me	(**1a**)	88 (**3a**)
2	*n*-Bu	H	(**1b**)	78 (**3b**)
3	*n*-Bu	Et	(**1c**)	81 (**3c**)
4	*n*-Bu	*n*-Pr	(**1d**)	84 (**3d**)
5	Me	Me	(**1e**)	64 (**3e**)
6	n-C_7H_{15}	Me	(**1f**)	84 (**3f**)
7	*t*-Bu	Me	(**1g**)	71 (**3g**)
8	Allyl	Me	(**1h**)	66 (**3h**)
9	Bn	Me	(**1i**)	89 (**3i**)
10	H	*n*-Pr	(**1j**)	51 (**3j**)
11	H	H	(**1k**)	31 [a](**3k**)

[a] The yield was determined by NMR using *p*-methoxyiodobenzene as the internal standard.

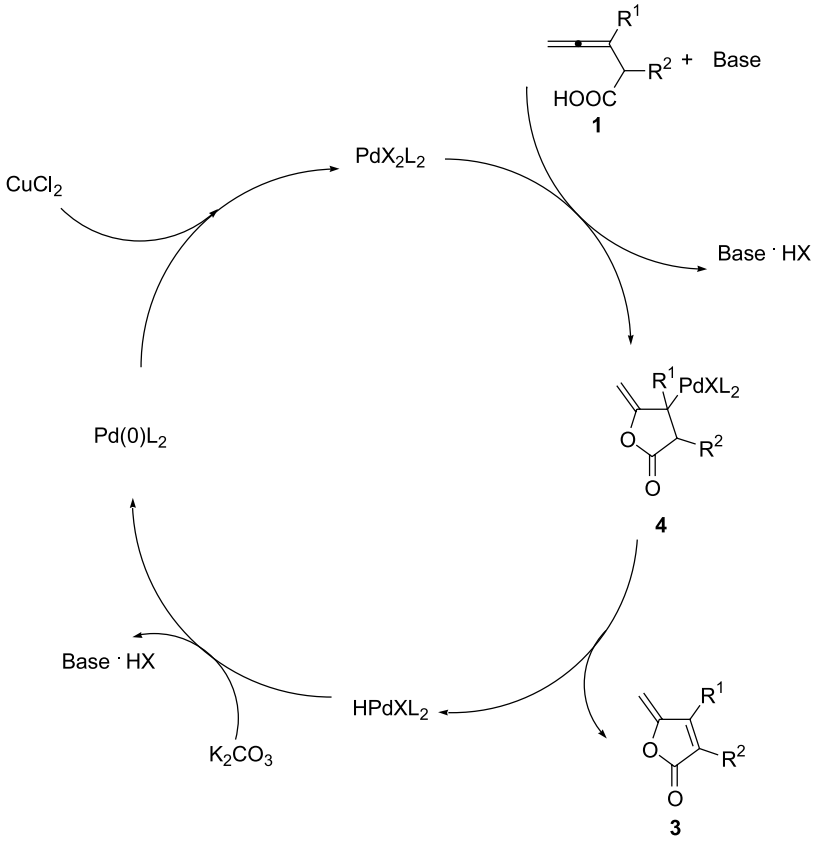
A palladium catalyzed mechanism was proposed for this reaction: *endo*-mode cyclic anti-oxypalladation of the non-terminal carbon–carbon double bonds in the 3,4-allenoic acids would form **4**, which would afford the products **3** via *syn*-β-H elimination, which may be facilitated by the presence of the base and the electron-withdrawing carbonyl group in **4**. The in situ generated palladium hydride species would form Pd(0) species upon its action with K_2CO_3. The catalytically active Pd(II) species would be regenerated via the oxidation with $CuCl_2$ (Scheme 3).

In conclusion, we have described a convenient method for the preparation of the valuable γ-methylene-2(5*H*)-furanones from 3,4-allenoic acids in high yield. The reaction demonstrated high efficiency and regioselectivity. Further studies in this area are being conducted in our laboratory.

3. Experimental

3.1. General procedure for the synthesis of β-allenoic acids 1

Conditions A. A solution of the corresponding β-allenoic acid ethyl ester[10] (6 mmol), and LiOH (288 mg, 12 mmol) was stirred in the mixed solvent of 12 mL of H_2O and 24 mL of MeOH at 30 °C for 3 days. After acidification with 1 N HCl, extraction with ether, drying over anhydrous Na_2SO_4, and evaporation, the mixture was submitted to flash



Scheme 3.

chromatography on silica gel (CH_2Cl_2/MeOH = 10/1) to produce compound **1**.

Conditions B. A solution of the corresponding β-allenoic acid ethyl ester[10] (6.6 mmol) was stirred in the mixed solvent of 27 mL of 20% hydrochloric acid, 20 mL of 1,4-dioxane, and 7 mL of THF at 30 °C for 3 days. After extraction with ether, drying over anhydrous Na_2SO_4, and evaporation, the mixture was submitted to flash chromatography on silica gel (CH_2Cl_2/MeOH = 10/1) to produce compound **1**.

3.1.1. 2-Methyl-3-(*n*-butyl)penta-3,4-dienoic acid (1a). The reaction of the corresponding allenoic acid ester (1044 mg, 5.3 mmol) with LiOH (250 mg, 10.4 mmol) afforded 817 mg (91%) of **1a** under conditions A: liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 11.64 (br s, 1H), 4.85–4.81 (m, 2H), 3.05–2.95 (m, 1H), 2.08–1.93 (m, 2H), 1.47–1.30 (m, 4H), 1.27 (d, J = 7.2 Hz, 3H), 0.89 (t, J = 7.2 Hz, 3H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 205.9, 181.3, 103.3, 78.3, 42.1, 30.2, 29.6, 22.3, 15.8, 13.9; IR (neat) 1958, 1711, 1459, 1222 cm^{-1}; MS (*m*/*z*) 168 (M^+, 4.83), 111 (100); HRMS calcd for $C_{10}H_{16}O_2(M^+)$ 168.11503; Found 168.11785.

3.1.2. 3-(*n*-Butyl)penta-3,4-dienoic acid (1b). The reaction of the corresponding allenoic acid ester (1.132 g, 6.2 mmol) with LiOH (314 mg, 13 mmol) afforded 275 mg (29%) of **1b** under conditions A: liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 10.83 (br s, 1H), 4.78–4.70 (m, 2H), 3.03 (t, J = 2.4 Hz, 2H), 2.08–1.99 (m, 2H), 1.48–1.25 (m, 4H), 0.89 (t, J = 7.2 Hz, 3H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 207.1, 178.0, 96.7, 76.1, 38.5, 31.2, 29.4, 22.2, 13.8; IR (neat) 1961, 1740 cm^{-1}; MS (*m*/*z*) 154 (M^+, 11.97), 41 (100); HRMS calcd for $C_9H_{14}O_2(M^+)$ 154.09938; Found 154.10058.

3.1.3. 2-Ethyl-3-(*n*-butyl)penta-3,4-dienoic acid (1c). The reaction of the corresponding allenoic acid ester (3654 mg, 17.4 mmol) with LiOH (945 mg, 39.4 mmol) afforded 3060 mg (97%) of **1c** under conditions A: liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 11.52 (br s, 1H), 4.84–4.79 (m, 2H), 2.77 (t, J = 7.5 Hz, 1H), 2.08–1.92 (m, 2H), 1.85–1.71 (m, 1H), 1.69–1.56 (m, 1H), 1.48–1.25 (m, 4H), 0.94 (t, J = 7.5 Hz, 3H), 0.89 (t, J = 7.2 Hz, 3H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 206.2, 180.7, 101.8, 77.8, 49.9, 30.3, 29.5, 23.5, 22.3, 13.9, 12.0; IR (neat) 1957, 1707, 1460, 1219 cm^{-1}; MS (*m*/*z*) 182 (M^+, 1.59), 153 (100); HRMS calcd for $C_{11}H_{18}O_2Na(MNa^+)$ 205.1223; Found 205.1219.

3.1.4. 2-*n*-(Propyl)-3-*n*-(butyl)penta-3,4-dienoic acid (1d). The reaction of the corresponding allenoic acid ester (4613 mg, 20.6 mmol) with LiOH (1198 mg, 50 mmol) afforded 3300 mg (82%) of **1d** under conditions A: liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 11.50 (br s, 1H), 4.84–4.76 (m, 2H), 2.87 (t, J = 7.5 Hz, 1H), 2.06–1.92 (m, 2H), 1.63–1.52 (m, 1H), 1.45–1.25 (m, 4H), 1.49–1.25 (m, 6H), 0.91 (t, J = 6.9 Hz, 3H), 0.89 (t, J = 7.2 Hz, 3H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 206.2, 180.9, 102.0, 77.8, 48.0, 32.4, 30.2, 29.5, 22.3, 20.6, 13.9, 13.8; IR (neat) 1957, 1708, 1466, 1210 cm^{-1}; MS (*m*/*z*) 196 (M^+, 1.19), 153 (100); HRMS calcd for $C_{12}H_{20}O_2Na(MNa^+)$ 219.1355; Found 219.1377.

3.1.5. 2,3-Dimethylpenta-3,4-dienoic acid (1e). The reaction of the corresponding allenoic acid ester (1.541 g, 10 mmol) with LiOH (480 mg, 20 mmol) afforded 639 mg (51%) of **1e** under conditions A: liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 11.77 (br s, 1H), 4.79–4.72 (m, 2H), 3.08–2.96 (m, 1H), 1.76 (t, J=3.3 Hz, 3H), 1.28 (d, J=6.9 Hz, 3H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 206.5, 181.1, 98.0, 76.3, 42.9, 17.1, 15.5; IR (neat) 1962, 1712, 1456, 1223 cm^{-1}; MS (*m*/*z*) 126 (M^+, 7.53), 111 (100); HRMS calcd for $C_7H_{10}O_2$(M^+) 126.06635; Found 126.06681.

3.1.6. 2-Methyl-3-(*n*-heptyl)penta-3,4-dienoic acid (1f). The reaction of the corresponding allenoic acid ester (1.357 g, 5.7 mmol) with LiOH (297 mg, 12.4 mmol) afforded 639 mg (53%) of **1f** under conditions A: liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 11.43 (br s, 1H), 4.86–4.81 (m, 2H), 3.08–2.95 (m, 1H), 2.14–1.92 (m, 2H), 1.48–1.26 (m, 13H), 0.88 (d, J=7.2 Hz, 3H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 205.9, 181.3, 103.4, 78.2, 42.1, 31.8, 30.5, 29.2, 29.1, 27.4, 22.6, 15.8, 14.1; IR (neat) 1958, 1709, 1459, 1221 cm^{-1}; MS (*m*/*z*) 210 (M^+, 3.75), 111 (100); HRMS calcd for $C_{13}H_{22}O_2$(M^+) 210.16198; Found 210.16407.

3.1.7. 2-Methyl-3-(*tert*-butyl)penta-3,4-dienoic acid (1g). The reaction of the corresponding allenoic acid ester (538 mg, 2.7 mmol) with LiOH (145 mg, 5.5 mmol) afforded 143 mg (31%) of **1g** under conditions A: solid; mp 44 °C (ether); ^{1}H NMR (300 MHz, $CDCl_3$) δ 4.86 (s, 2H), 3.06 (q, J=6.9 Hz, 1H), 1.29 (d, J=7.2 Hz, 3H), 1.08 (s, 9H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 205.3, 181.6, 112.8, 79.1, 37.6, 33.6, 29.0, 18.4; IR (neat) 1950, 1707, 1233 cm^{-1}; MS (*m*/*z*) 169 (M^++1, 93.02), 168 (M^+, 20.46), 153 (100). Anal. Calcd for $C_{10}H_{16}O_2$: C, 71.39; H, 9.59. Found C, 71.46; H, 9.31.

3.1.8. 2-Methyl-3-allylpenta-3,4-dienoic acid (1h). The reaction of the corresponding allenoic acid ester (2745 mg, 15.2 mmol) with LiOH (719 mg, 30 mmol) afforded 553 mg (24%) of **1h** under conditions A: liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 10.57 (br s, 1H), 5.88–5.71 (m, 1H), 5.12–5.02 (m, 2H), 4.88–4.83 (m, 2H), 3.09–2.98 (m, 1H), 2.90–2.73 (m, 2H), 1.28 (d, J=6.6 Hz, 3H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 206.3, 180.9, 135.1, 116.5, 101.6, 78.3, 41.2, 35.7, 15.7; IR (neat) 1958, 1712, 1641 cm^{-1}; MS (*m*/*z*) 153 (M^++1, 38.90), 152 (M^+, 14.13), 107 (100); HRMS calcd for $C_9H_{12}O_2$(M^+) 152.08373; Found 152.08326.

3.1.9. 2-Methyl-3-benzylpenta-3,4-dienoic acid (1i). The reaction of the corresponding allenoic acid ester (1.798 g, 7.8 mmol) with LiOH (402 mg, 16.8 mmol) afforded 1.179 g (75%) of **1i** under conditions A: liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 11.78 (br s, 1H), 7.38–7.15 (m, 5H), 4.87 (br s, 2H), 3.52 (d, J=15.0 Hz, 1H), 3.41 (d, J=15.0 Hz, 1H), 3.03–2.92 (m, 1H), 1.28 (d, J=7.2 Hz, 3H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 206.9, 181.1, 138.7, 129.1, 128.2, 126.4, 102.9, 78.1, 40.6, 38.2, 15.9; IR (neat) 1958, 1705, 1455, 1226 cm^{-1}; MS (*m*/*z*) 202 (M^+, 2.59), 129 (100); HRMS calcd for $C_{13}H_{14}O_2$(M^+) 202.09938; Found 202.09801.

3.1.10. 2-(*n*-Propyl)penta-3,4-dienoic acid (1j). The reaction of the corresponding allenoic acid ester (1693 mg, 10 mmol) afforded 453 mg (33%) of **1j** under conditions B: liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 11.60 (br s, 1H), 5.20 (q, J=3.6 Hz, 1H), 4.80 (d, J=6.9 Hz, 2H), 3.03 (q, J=8.1 Hz, 1H), 1.83–1.72 (m, 1H), 1.64–1.55 (m, 1H), 1.43–1.34 (m, 2H), 0.92 (t, J=7.5 Hz, 3H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 208.5, 180.8, 88.7, 76.5, 44.9, 33.9, 20.2, 13.7; IR (neat) 1958, 1709, 1415, 1286, 1208 cm^{-1}; MS (*m*/*z*) 141 (M^++1, 4.15), 97 (100); HRMS calcd for $C_8H_{11}O$(M^+−OH) 123.08099; Found 123.08177.

3.1.11. Penta-3,4-dienoic acid (1k).[11] The reaction of the corresponding allenoic acid ester (2038 mg, 16.2 mmol) afforded 1360 mg (86%) of **1k** under conditions B: liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 11.91 (br s, 1H), 5.23 (pentet, J=6.9 Hz, 1H), 4.79–4.71 (m, 2H), 3.12–3.04 (m, 2H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 209.3, 178.1, 82.7, 75.9, 33.9; IR (neat) 1960, 1713, 1440 cm^{-1}; MS (*m*/*z*) 98 (M^+, 15.03), 97 (M^+−1, 40.04), 70 (100).

The reaction of **1a** (86 mg, 5.0 mmol), PhI (85 μl, 7.5 mmol), K_2CO_3 (273 mg, 2.0 mmol), $AgNO_3$ (8 mg, 0.05 mmol), and $Pd(PPh_3)_4$ (30 mg, 0.025 mmol) in 3 mL of CH_3CN at 70 °C for 33 h produced 30 mg (24%) of **2a** and 9 mg (11%) of **3a** via separation with chromatography on silica gel (petroleum ether/ether=5/1).

3.1.12. 3-Methyl-4-(*n*-butyl)-5-benzyl-2(5*H*)-furanone (2a). Liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.32–7.17 (m, 5H), 5.09–4.97 (m, 1H), 3.15 (dd, J=3.6, 14.1 Hz, 1H), 2.75 (dd, J=7.5, 14.1 Hz, 1H), 2.53–2.40 (m, 1H), 2.30–2.22 (m, 1H), 1.74 (s, 3H), 1.52–1.22 (m, 4H), 0.92 (t, J=7.2 Hz, 3H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 174.3, 162.5, 135.6, 129.2, 128.3, 126.8, 123.9, 82.1, 38.6, 29.7, 26.4, 22.5, 13.6, 8.5; IR (neat) 1754, 1675, 1455 cm^{-1}; MS (*m*/*z*) 244 (M^+, 0.62), 84 (100); HRMS calcd for $C_{16}H_{20}O_2$(M^+) 244.14633; Found 244.14523.

3.2. General procedure for $PdCl_2(PPh_3)_2$-catalyzed cyclization reaction of β-alleneoic acids 1

Under N_2 atmosphere a solution of **1** (0.5 mmol), $PdCl_2(PPh_3)_2$ (18 mg, 5 mol%), anhydrous $CuCl_2$ (270 mg, 2 mmol), and K_2CO_3 (276 mg, 2 mmol) was stirred in 3 mL of anhydrous DMF at 70 °C for 12 h. The reaction mixture was diluted with ether (50 mL), washed with brine, and dried over anhydrous Na_2SO_4. After evaporation, the residue was submitted to column chromatography on silica gel (petroleum ether/ether=10/1) to produce **3**.

3.2.1. 3-Methyl-4-(*n*-butyl)-5-methylene-2(5*H*)-furanone (3a). The reaction of **1a** (84 mg, 0.5 mmol), $CuCl_2$ (266 mg, 2.0 mmol), K_2CO_3 (277 mg, 2.0 mmol), and $Pd(PPh_3)_4$ (5 mol%) in 3 mL of DMF at 70 °C for 12 h produced 73 mg (88%) of **3a**: liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 5.06 (d, J=2.7 Hz, 1H), 4.81 (d, J=2.7 Hz, 1H), 2.44 (t, J=7.5 Hz, 2H), 1.91 (s, 3H), 1.56–1.46 (m, 2H), 1.41–1.30 (m, 2H), 0.93 (t, J=7.2 Hz, 3H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 170.7, 155.2, 150.9, 126.0, 92.5, 31.1, 24.5, 22.6, 13.8, 8.8; IR (neat) 1773, 1648, 1459, 1287 cm^{-1}; MS (*m*/*z*) 167 (M^++1, 10.40), 166 (M^+, 10.40), 124 (100); HRMS calcd for $C_9H_{11}O_2$(M^+−CH_3) 151.0759; Found 151.0785.

3.2.2. 5-Methylene-4-(*n*-butyl)-2(5*H*)-furanone (3b). The reaction of **1b** (74 mg, 0.5 mmol), $CuCl_2$ (267 mg, 2.0 mmol), K_2CO_3 (276 mg, 2.0 mmol), and $Pd(PPh_3)_4$ (5 mol%) in 3 mL of DMF at 70 °C for 12 h produced 57 mg (78%) of **3b**: liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 5.95–5.92 (m, 1H), 5.13–5.10 (m, 1H), 4.91–4.89 (m, 1H), 2.45 (dt, J=1.2, 7.5 Hz, 2H), 1.66–1.52 (m, 2H), 1.46–1.31 (m, 2H), 0.92 (t, J=7.5 Hz, 3H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 169.2, 159.3, 155.9, 116.6, 94.1, 29.9, 25.7, 22.2, 13.6; IR (neat) 1763, 1651, 1467, 1261 cm^{-1}; MS (*m*/*z*) 153 (M^++1, 21.12), 152 (M^+, 12.22), 110 (100); HRMS calcd for $C_9H_{12}O_2(M^+)$ 152.08373; Found 152.08498.

3.2.3. 3-Ethyl-4-(*n*-butyl)-5-methylene-2(5*H*)-furanone (3c). The reaction of **1c** (92 mg, 0.5 mmol), $CuCl_2$ (282 mg, 2.0 mmol), K_2CO_3 (286 mg, 2.0 mmol), and $Pd(PPh_3)_4$ (5 mol%) in 3 mL of DMF at 70 °C for 12 h produced 74 mg (81%) of **3c**: liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 5.04 (d, J=2.7 Hz, 1H), 4.79 (d, J=2.4 Hz, 1H), 2.42 (t, J=7.5 Hz, 2H), 2.32 (q, J=7.5 Hz, 2H), 1.57–1.46 (m, 2H), 1.44–1.30 (m, 2H), 1.10 (t, J=7.5 Hz, 3H), 0.91 (t, J=6.9 Hz, 3H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 170.1, 155.1, 150.4, 131.3, 92.5, 31.7, 24.3, 22.6, 17.2, 13.7, 12.7; IR (neat) 1772, 1648, 1463, 1296, 1256 cm^{-1}; MS (*m*/*z*) 181 (M^++1, 100); HRMS calcd for $C_{11}H_{16}O_2Na(MNa)^+$ 203.1042; Found 203.1041.

3.2.4. 3-(*n*-Propyl)-4-(*n*-butyl)-5-methylene-2(5*H*)-furanone (3d). The reaction of **1d** (98 mg, 0.5 mmol), $CuCl_2$ (272 mg, 2.0 mmol), K_2CO_3 (278 mg, 2.0 mmol), and $Pd(PPh_3)_4$ (5 mol%) in 3 mL of DMF at 70 °C for 12 h produced 81 mg (84%) of **3d**: liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 5.03 (d, J=2.4 Hz, 1H), 4.79 (d, J=2.7 Hz, 1H), 2.41 (t, J=7.5 Hz, 2H), 2.26 (t, J=7.5 Hz, 2H), 1.58–1.46 (m, 4H), 1.40–1.30 (m, 2H), 0.91 (t, J=7.5 Hz, 6H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 170.2, 155.0, 150.9, 129.9, 92.5, 31.6, 25.7, 24.4, 22.7, 21.4, 13.9, 13.7; IR (neat) 1718, 1648, 1465, 1284 cm^{-1}; MS (*m*/*z*) 195 (M^++1, 32.35), 194 (M^+, 15.12), 123 (100); HRMS calcd for $C_{12}H_{18}O_2(M^+)$ 194.13068; Found 194.13200.

3.2.5. 3,4-Dimethyl-5-methylene-2(5*H*)-furanone (3e). The reaction of **1e** (63 mg, 0.5 mmol), $CuCl_2$ (270 mg, 2.0 mmol), K_2CO_3 (281 mg, 2.0 mmol), and $Pd(PPh_3)_4$ (5 mol%) in 3 mL of DMF at 70 °C for 12 h produced 40 mg (64%) of **3e**: solid; mp 80–81 °C (ether); ^{1}H NMR (300 MHz, $CDCl_3$) δ 5.03 (d, J=2.7 Hz, 1H), 4.79 (d, J=2.7 Hz, 1H), 2.04 (s, 3H), 1.89 (s, 3H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 170.6, 155.8, 146.6, 126.2, 92.2, 9.9, 8.7; IR (neat) 1763, 1651, 1439, 1285, 1260 cm^{-1}; MS (*m*/*z*) 124 (M^+, 100); HRMS calcd for $C_7H_8O_2(M^+)$ 124.05243; Found 124.05262.

3.2.6. 3-Methyl-4-(*n*-heptyl)-5-methylene-2(5*H*)-furanone (3f). The reaction of **1f** (105 mg, 0.5 mmol), $CuCl_2$ (277 mg, 2.0 mmol), K_2CO_3 (282 mg, 2.0 mmol), and $Pd(PPh_3)_4$ (5 mol%) in 3 mL of DMF at 70 °C for 12 h produced 90 mg (84%) of **3f**: liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 5.03 (d, J=2.1 Hz, 1H), 4.78 (d, J=2.7 Hz, 1H), 2.40 (t, J=7.5 Hz, 2H), 1.88 (s, 3H), 1.53–1.46 (m, 2H), 1.30–1.21 (m, 8H), 0.84 (t, J=6.9 Hz, 3H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 170.7, 155.2, 151.0, 126.0, 92.5, 31.6, 29.4, 29.0, 28.9, 24.8, 22.5, 14.0, 8.8; IR (neat) 1774, 1648, 1466, 1287 cm^{-1}; MS (*m*/*z*) 208 (M^+, 6.83), 124 (100); HRMS calcd for $C_{13}H_{20}O_2(M^+)$ 208.14633; Found 208.14484.

3.2.7. 3-Methyl-4-(*tert*-butyl)-5-methylene-2(5*H*)-furanone (3g). The reaction of **1g** (77 mg, 0.46 mmol), $CuCl_2$ (250 mg, 1.85 mmol), K_2CO_3 (262 mg, 1.90 mmol), and $Pd(PPh_3)_4$ (5 mol%) in 3 mL of DMF at 70 °C for 12 h produced 53 mg (71%) of **3g**: liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 5.21 (d, J=2.1 Hz, 1H), 5.09 (d, J=2.1 Hz, 1H), 2.08 (s, 3H), 1.39 (s, 9H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 170.9, 155.5, 154.0, 125.4, 97.0, 34.7, 30.9, 11.3; IR (neat) 1770, 1639, 1464, 1274 cm^{-1}; MS (*m*/*z*) 166 (M^+, 59.06), 123 (100); HRMS calcd for $C_{10}H_{14}O_2Na(MNa)^+$189.0886; Found 189.0894.

3.2.8. 3-Methyl-4-allyl-5-methylene-2(5*H*)-furanone (3h). The reaction of **1h** (77 mg, 0.5 mmol), $CuCl_2$ (269 mg, 2.0 mmol), K_2CO_3 (280 mg, 2.0 mmol), and $Pd(PPh_3)_4$ (5 mol%) in 3 mL of DMF at 70 °C for 12 h produced 50 mg (66%) of **3h**: liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 5.86–5.72 (m, 1H), 5.15–5.08 (m, 3H), 4.84 (d, J=3.0 Hz, 1H), 3.21 (d, J=6.0 Hz, 2H), 1.92 (s, 3H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 170.4, 154.8, 147.7, 132.4, 127.1, 117.5, 93.1, 28.9, 8.8; IR (neat) 1782, 1650, 1441, 1287 cm^{-1}; MS (*m*/*z*) 150 (M^+, 36.37), 79 (100); HRMS calcd for $C_9H_{10}O_2Na(MNa^+)$ 173.0573; Found 173.0580.

3.2.9. 3-Methyl-4-benzyl-5-methylene-2(5*H*)-furanone (3i). The reaction of **1i** (101 mg, 0.5 mmol), $CuCl_2$ (264 mg, 2.0 mmol), K_2CO_3 (282 mg, 2.0 mmol), and $Pd(PPh_3)_4$ (5 mol%) in 3 mL of DMF at 70 °C for 12 h produced 90 mg (89%) of **3i**: liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.32–7.13 (m, 5H), 5.08 (d, J=2.7 Hz, 1H), 4.81 (d, J=2.7 Hz, 1H), 3.82 (s, 2H), 1.94 (s, 3H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 170.3, 154.9, 148.4, 136.5, 128.8, 128.1, 127.4, 126.9, 93.7, 30.6, 9.0; IR (neat) 1770, 1648, 1496, 1454, 1287 cm^{-1}; MS (*m*/*z*) 200 (M^+, 41.10), 84 (100); HRMS calcd for $C_{13}H_{12}O_2Na(MNa^+)$ 223.0729; Found 223.0737.

3.2.10. 3-(*n*-Propyl)-5-methylene-2(5*H*)-furanone (3j). The reaction of **1j** (70 mg, 0.5 mmol), $CuCl_2$ (270 mg, 2.0 mmol), K_2CO_3 (286 mg, 2.0 mmol), and $Pd(PPh_3)_4$ (5 mol%) in 3 mL of DMF at 70 °C for 12 h produced 36 mg (51%) of **3j**: liquid; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.02 (s, 1H), 5.08 (dd, J=1.2, 2.4 Hz, 1H), 4.76 (d, J=2.4 Hz, 1H), 2.38–2.28 (m, 2H), 1.66–1.54 (m, 2H), 0.95 (t, J=6.9 Hz, 3H); ^{13}C NMR (75.4 MHz, $CDCl_3$) δ 170.5, 153.9, 136.4, 128.5, 96.5, 27.1, 20.7, 13.6; IR (neat) 1767, 1465 cm^{-1}; MS (*m*/*z*) 138 (M^+, 24.48), 84 (100); HRMS calcd for $C_8H_{10}O_2Na(MNa)^+$ 161.0573; Found 161.0581.

3.2.11. 5-Methylene-2(5*H*)-furanone (3k).[12] The reaction of **1k** (98 mg, 1.0 mmol), $CuCl_2$ (545 mg, 4.0 mmol), K_2CO_3 (550 mg, 4.0 mmol), and $Pd(PPh_3)_4$ (5 mol%) in 3 mL of DMF at 70 °C for 12 h produced **3k** (31%) (determined by NMR using *p*-methoxyiodobenzene as the internal standard).

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Corrigendum

Corrigendum to "A palladium catalyzed efficient synthesis of γ-methylene-α,β-unsaturated γ-lactones via cyclization of 3,4-alkadienoic acids" [Tetrahedron 61 (2005) 9896]

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In Sections 3.2.1–3.2.11 of the paper '$Pd(PPh_3)_4$' should be read as '$PdCl_2(PPh_3)_2$'.

The authors wish to apologize for this error.

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